ANALYSTLAB AFRICA

Data Science Internship Programme

Heart Disease Prediction
Feature Evaluation and Selection Summary Report

By

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Project: Heart Disease Prediction
Dataset: Heart Disease Prediction Dataset

Prepared for the AnalystLab Africa Data Science Internship Programme
Week 3 Submission – Advanced Data Analysis, Statistical Testing & Feature Engineering

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1. Introduction

This report presents the Feature Evaluation and Selection Summary for the Heart Disease Prediction project, prepared as part of the AnalystLab Africa Data Science Internship Programme, Week 3: Advanced Data Analysis, Statistical Validation & Feature Engineering.

The Week 3 brief requires that both original and newly engineered features be evaluated to determine which variables provide useful information for future modelling. Specifically, the evaluation must consider correlation between numerical variables, highly correlated or redundant variables, potential multicollinearity, features with limited analytical value, the relationship between each feature and the target variable, and potential data leakage. The brief is explicit that a feature should not be removed simply because it is correlated with another; every retention or removal decision must be justified with evidence.

This report draws on the statistical evidence generated during the Week 3 Advanced Exploratory Data Analysis and the accompanying Statistical Analysis Summary, which established that Age, MaxHR, Oldpeak, ST_Slope, Exercise Angina and Chest Pain Type are all significantly associated with heart disease status. Building on that evidence, this report evaluates the full Final_Modelling_Dataset, comprising the original clinical variables and six newly engineered features, against each of the criteria set out in the Week 3 brief, and sets out clear retention decisions for the dataset that will be carried forward into Week 4 Machine Learning.

2. Dataset Structure Overview

The Final_Modelling_Dataset contains 26 columns in total, composed of 20 original features carried over from the clean dataset produced in Week 2, and 6 newly engineered features added during Week 3 to enhance predictive power or interpretability.

Three of the original variables (FastingBS, Sex_M, ExerciseAngina_Y) were transformed in place from their numerical or boolean representation into readable categorical string labels, to make the Week 3 exploratory and statistical analysis easier to interpret.

A discrepancy was identified while reviewing the feature documentation for this report: although the underlying notes describe **six** newly engineered features, seven are individually listed — ChestPainType, RestingECG, ST_Slope, Age_Group, BP_Category, Heart_Rate_Age_Ratio, and ST_Exercise_Risk. On closer inspection, ChestPainType, RestingECG and ST_Slope are human-readable reconstructions of categorical variables that already existed in the dataset in one-hot encoded form from Week 2; they make the analysis easier to read but do not introduce new predictive information. Age_Group, BP_Category, Heart_Rate_Age_Ratio and ST_Exercise_Risk are the four variables that represent genuinely new engineered information. This distinction is addressed further in Section 5 and should be clarified in the final feature documentation.

3. Evaluation Framework

Consistent with the Week 3 brief, each feature (or group of related features) was evaluated against four criteria:

- **Relationship with the target variable:** whether the feature shows a statistically significant or practically meaningful association with HeartDisease.
- **Multicollinearity and redundancy:** whether the feature duplicates information already captured by another feature.
- **Analytical value:** whether the feature contributes a meaningful, interpretable signal, even where that signal is comparatively weak.

- **Data leakage:** whether the feature could have been influenced by, or could only be known after, the outcome it is intended to help predict (Kaufman et al., 2012).

For each criterion, a feature was retained unless there was clear evidence that removing it would improve, or at least not harm, the quality and interpretability of the final modelling dataset. No feature was removed on the basis of correlation alone, consistent with the Week 3 brief's instruction that correlation by itself is not sufficient grounds for removal.

4. Relationship with the Target Variable (HeartDisease)

Evidence: The Statistical Analysis Summary prepared earlier in Week 3 tested the relationship between HeartDisease and each of the key clinical variables:

- Age, MaxHR and Oldpeak all showed statistically significant differences between the heart disease groups (Mann-Whitney U tests, $p < 0.001$).
- ST_Slope, Exercise Angina and Chest Pain Type all showed strong, statistically significant associations with heart disease (Chi-Square tests, $p < 0.001$).
- FastingBS also showed a significant association with heart disease.

The engineered features showed similarly meaningful patterns:

- **Age_Group:** Heart disease prevalence increased clearly with age group, rising to 73% among patients classified as 'Elderly'.
- **BP_Category:** Patients classified as 'Hypertension' showed a higher heart disease prevalence of 63%.
- **Heart_Rate_Age_Ratio:** Showed a moderate negative correlation with HeartDisease ($r = -0.403$), with the boxplot showing clearly lower ratios in the heart disease group. This feature condenses the combined predictive information of Age and MaxHR into a single variable.
- **ST_Exercise_Risk:** A cross-tabulation of this ST_Slope / Exercise Angina interaction revealed very strong associations, with combinations such as 'Flat_Yes' and 'Down_Yes' showing an especially high prevalence of heart disease.

Decision: All original and engineered features listed above are retained as potentially useful for modelling.

Justification: Features that show a statistically significant relationship or a meaningful correlation with the target variable generally help explain variability in the outcome and are valuable for predictive modelling. Even features with weaker individual associations, such as RestingBP and Cholesterol, are retained at this stage due to their established clinical relevance and the possibility that their effects become more apparent in combination with other variables in a multivariate model.

5. Multicollinearity and Redundancy

Evidence: The Spearman correlation analysis conducted during the Week 3 statistical testing indicated mostly weak to moderate relationships among the numerical predictor variables. A systematic check for highly correlated pairs (using a threshold of $|r| > 0.80$) confirmed that no pair of numerical variables exceeded this threshold.

Decision: No numerical features are removed on the grounds of multicollinearity.

Justification: The absence of strong multicollinearity ($|r| > 0.80$) among the numerical features is beneficial: it suggests these variables carry largely independent information, which supports stable and interpretable coefficients in linear models and is generally unproblematic for tree-based methods (James et al., 2021).

A separate, more specific redundancy concern was identified in this review, however, and is not addressed in the original notes: as noted in Section 2, ChestPainType, RestingECG and ST_Slope are human-readable reconstructions of the same information already present in the dataset as one-hot encoded columns from Week 2. If both the reconstructed categorical column and its corresponding one-hot encoded columns are fed into the same model, the same clinical information would be represented twice, which is a form of redundancy rather than multicollinearity in the conventional numerical sense.

Recommendation: Before Week 4 modelling begins, the Final_Modelling_Dataset should retain either the readable categorical version of ChestPainType, RestingECG and ST_Slope, or their corresponding one-hot encoded columns, but not both, for whichever representation the chosen modelling algorithm requires.

6. Features with Limited Analytical Value

Evidence: While all features showed some association with HeartDisease, RestingBP and Cholesterol showed less clear visual separation in their boxplots by HeartDisease status and comparatively weaker correlations with HeartDisease in the Spearman correlation matrix. Both are, however, clinically recognised cardiovascular risk factors.

Decision: Both features are retained.

Justification: No feature was judged to have limited analytical value to the extent of justifying removal. Features with weaker univariate associations can still contribute meaningfully in a multivariate context or through interaction effects with other variables (Kuhn & Johnson, 2019). Given their established medical importance, RestingBP and Cholesterol are considered important to retain for a clinically comprehensive model, despite their comparatively modest individual signal within this particular dataset.

7. Potential Data Leakage

The Week 3 brief requires that the dataset also be checked for potential data leakage, that is, for the unintentional inclusion of information about the target variable that would not legitimately be available at the point a prediction is needed (Kaufman et al., 2012). This check was not documented in the original feature evaluation notes and is added here.

Evidence: All 26 columns in the Final_Modelling_Dataset represent patient characteristics, symptoms or diagnostic test results that would ordinarily be recorded during a standard clinical work-up, before or at the point a diagnosis of heart disease is confirmed. None of the variables (including the engineered Age_Group, BP_Category, Heart_Rate_Age_Ratio and ST_Exercise_Risk) are derived from, or could only be known after, the HeartDisease outcome itself. No post-diagnosis, treatment-related or outcome-adjacent variables (for example, medication records or follow-up test results) are present in the dataset.

Decision: No features are removed on the grounds of data leakage.

Justification: Because every feature reflects information that would plausibly be available prior to, or at the point of, diagnosis, there is no evidence of target leakage in the Final_Modelling_Dataset. This should nonetheless be revisited in Week 4 if any additional features are engineered, and it should be confirmed that any scaling, encoding or imputation statistics used in preprocessing are fitted only on training data once the dataset is split, to avoid introducing leakage at the modelling stage (Kaufman et al., 2012).

8. Summary of Retention Decisions

The table below summarises the evaluation and decision for the features discussed in this report.

Feature	Type	Key Evidence	Decision
Age, MaxHR, Oldpeak	Original (numerical)	Significant group difference vs. HeartDisease (Mann-Whitney U, $p < 0.001$)	Retain
ST_Slope, ExerciseAngina, ChestPainType	Original (categorical)	Significant association with HeartDisease (Chi-Square, $p < 0.001$)	Retain
FastingBS	Original (binary)	Significant association with HeartDisease	Retain
RestingBP, Cholesterol	Original (numerical)	Weaker individual signal; established clinical relevance	Retain
Age_Group	Engineered (categorical)	HeartDisease prevalence rises to 73% for 'Elderly'	Retain
BP_Category	Engineered (categorical)	HeartDisease prevalence of 63% for 'Hypertension'	Retain
Heart_Rate_Age_Ratio	Engineered (numerical)	Moderate correlation with HeartDisease ($r = -0.403$)	Retain
ST_Exercise_Risk	Engineered (interaction)	Very high HeartDisease prevalence for specific combinations	Retain
ChestPainType / RestingECG / ST_Slope (reconstructed) vs. their one-hot columns	Redundancy check	Same information represented twice if both kept	Retain one representation only
All numerical features (pairwise)	Multicollinearity check	No pair exceeded $\|r\| > 0.80$	No removal required
All 26 features	Data leakage check	All variables plausibly available pre-diagnosis	No removal required

9. Overall Conclusion

Based on this evaluation, all original and newly engineered features in the Final_Modelling_Dataset are considered valuable for subsequent modelling. They exhibit meaningful relationships with heart disease, show limited multicollinearity among the numerical predictors, align with established clinical understanding, and show no evidence of data leakage.

The newly engineered features, in particular Heart_Rate_Age_Ratio and ST_Exercise_Risk, appear to capture combined and interaction effects that are not fully visible when the underlying variables are considered individually, and are expected to add genuine value in Week 4. Age_Group and BP_Category provide a more interpretable, clinically framed view of Age and RestingBP that may be useful for communicating results to non-technical stakeholders, even where the underlying continuous variable is retained for modelling.

The one outstanding action identified in this review is to resolve the duplicate representation of ChestPainType, RestingECG and ST_Slope (Section 5) before the dataset is finalised for Week 4, so that the same clinical information is not represented twice within the same model.

10. Limitations and Recommendations for Future Work

- **Correlation-based methods only:** This evaluation relied on Spearman correlation and the Week 3 hypothesis tests. The Week 3 brief also permits mutual information and feature importance from a preliminary model as evaluation methods; neither was used here. Running a simple preliminary model (for example, a logistic regression or a tree-based model) in Week 4 would provide an additional, model-based check on which features carry the most predictive weight.
- **No formal variance analysis:** A check for near-zero-variance features (variables with almost no variation across patients) was not performed. This is a quick, low-cost check that could be added before Week 4 modelling to rule out any feature that provides negligible information.
- **Dataset size:** As noted in the Week 3 Project Continuity Summary, the dataset contains 918 observations. This is a reasonable basis for the exploratory and statistical work carried out so far, but is a modest sample size for training and validating a machine-learning model with confidence, particularly once the dataset is split into training and test sets in Week 4.
- **Feature duplication:** As discussed in Section 5, the dataset currently contains both a readable categorical version and a one-hot encoded version of ChestPainType, RestingECG and ST_Slope. This should be resolved before modelling to avoid duplicated information.

References

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